

PROJECT PROPOSAL

This project investigates how different product characteristics influence user satisfaction in skincare products, focusing on the question: how do price, ingredient safety score, vegan/cruelty-free labeling, and product category affect user ratings? The main hypothesis is that ingredient safety has a stronger effect on user satisfaction than price, while the null hypothesis assumes no significant relationship between these variables and ratings. The project will use multiple real-world datasets, including the following sources:

- Sephora products and reviews dataset (Kaggle)
- Skincare ingredients dataset (Kaggle)
- EWG Skin Deep database
- Open Beauty Facts

Data will be collected by downloading datasets from Kaggle and retrieving additional information from public databases, then processed through several steps such as merging product and review data using product identifiers, parsing and standardizing ingredient lists, matching ingredients with safety scores, and generating new features such as overall safety score and vegan labeling. The final dataset is expected to include several thousand products and a large number of user reviews, with numerical variables (price, rating, safety score) and categorical variables (product category, vegan label). This structured dataset will enable the use of statistical analysis and regression models to evaluate the relative importance of each factor and test the proposed hypothesis.